

CS407 Project Charter

Team:

- Xiuwen(Charlie) Fang: fang331@purdue.edu
- Xinyi Guan: guan90@purdue.edu
- Yuchuan Huang: huan1695@purdue.edu
- Xinyi Wu: wu1765@purdue.edu

Problem Statement:

We aim to develop a beginner-friendly programming game that teaches fundamental coding concepts through a series of engaging, puzzle-based challenges designed for users with no prior experience in programming. Our project stands out by focusing on a story-driven, gamified learning experience, integrating problem-solving scenarios and a customizable code editor that adapts its complexity as the player progresses, ensuring a smooth learning curve. The game will consist of multiple levels, each introducing a specific programming concept such as loops, conditionals, or variables, with challenges that grow progressively more complex. Players will interact with a visually intuitive interface where they solve puzzles by writing or manipulating code blocks, simulating real-world programming tasks. To maintain engagement, the game will feature a storyline where players guide a character through obstacles using code, unlocking rewards and achievements as they advance. By combining hands-on practice with immediate feedback, the game ensures an effective and enjoyable learning experience for beginners.

Project Objectives:

- Develop a Role-Playing Game (RPG):
 - Core Concept: Creates an engaging role-playing game that combines basic programming education with interactive, classical mini-games for practice.
 - Progression Mechanics: Allows players to learn programming concepts step by step, with challenges and rewards tied to their progress.
 - Interactivity: Includes quizzes, puzzles, and coding challenges that adapt to the player's skill level.
- Story Background and Narrative:
 - Immersive Storyline: Crafts a compelling narrative that ties programming challenges to an adventurous plot.
 - Character Development: Enables players to collect experience or money to show their learning level of the topics.
 - Theme Integration: Links programming topics to the story.
- Map-Based Chapter Selection:
 - Visual Chapter Map: Creates a visually appealing, interactive map where each chapter represents a new programming concept.
 - Unlockable Areas: Designs the map so that players unlock new regions as they progress, with increasing difficulty levels.
 - Mini-Games Integration: Associates each map area with thematic mini-games
- Database Design and Implementation:

- Player Information: Stores details such as player profiles, levels, and achievements.
- Learning Records: Tracks progress on programming topics and completed challenges.

Stakeholders:

- Users/Customers: Young students who want to learn about programming logic and have their first try on some programming practice in a specific language.
- Project Owners: Xiuwen(Charlie) Fang, Xinyi Guan, Yuchuan Huang, Xinyi Wu
- Software Developers: Xiuwen(Charlie) Fang, Xinyi Guan, Yuchuan Huang, Xinyi Wu
- Development Manager: TBD (GTA unassigned)

Deliverables:

- A role-playing game created with **Unity**:
 - The game will feature a series of levels, each designed to teach specific programming concepts (e.g., loops, conditionals, functions). Players will progress through these levels by solving challenges integrated into the storyline.
- Level-specific content tailored to programming functions:
 - Each game level will introduce and reinforce a specific programming concept through interactive puzzles, challenges, or mini-games. These levels will align with educational goals and ensure incremental learning.
- A backend developed using **Node.js**, integrated with an **MYSQL** database:
 - The backend will manage user authentication, save each player's progress. It will ensure that all user data is stored securely and can be retrieved without loss.
- Cloud environment using **AWS**:
 - To support the role-playing game's scalability, performance, and reliability, AWS will be utilized as the primary cloud platform.

About CS307 Project:

- Xiuwen(Charlie) Fang, Xinyi Guan, Xinyi Wu:
 - Github Repository: <https://github.com/XWnina/foo-my-food>
 - Descriptions: In CS307, our group created an application called FooMyFood. This app is a comprehensive food management solution designed to address the challenges faced by individuals living alone. It enables users to track their food inventory by recording purchase and expiration dates, while sending timely reminders to use ingredients before they spoil, helping to minimize food waste and reduce financial burden. The app also offers personalized recipe suggestions tailored to the ingredients users already have, making meal planning easier and more creative. Additionally, it features a platform for users to create and share unique recipes, promoting culinary exploration, healthier eating habits, and greater control over dietary choices. By integrating food tracking, meal planning, and recipe sharing into one user-friendly platform,

the app simplifies daily food management and encourages smarter, more sustainable consumption.

- Yuchuan Huang:
 - Description: fathomless.io is a roguelike game with procedurally generated dungeons. Our project will combine several uncommon features seldom found within this general genre of game. The main goal of our development is to fix undesired mechanics encountered in other roguelike games while supplying a competitive, score based leaderboard. These newly implemented gameplay features will yield a series of fun, engaging, and challenging single player levels. In our single player game, the player controls a character with 4 way movement across progressively more complex level maps (starting at a simple ground level, going deeper into a cave dungeon, and ending at the elaborate “Underworld”). The player picks up items along the way that help them defeat monsters with the ultimate task of traversing/beating all the maze-like cave levels and killing a final boss.